

Polygamma Values at Complex Multiplication Points: Eisenstein Row-Sums, Chowla–Selberg Periods, and Hilbert Class Polynomials

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Abstract

Summing an Eisenstein series of the lattice $\mathbb{Z} + \mathbb{Z}\tau$ row by row turns it into a series of polygamma values at the non-real points $n\tau$. At complex multiplication (CM) points the Eisenstein values are pinned by classical theory, and the row-sum identity converts that information into explicit evaluations of infinite sums of polygamma values at complex arguments. We record three layers of consequences, all verified numerically at 50–300 digits with PSLQ-canary discipline. First, *vanishing identities*: at the square and hexagonal lattices the missing modular forms collapse polygamma sums to rational multiples of $\zeta(4)$ and $\zeta(6)$. Second, *period identities*: the surviving Eisenstein values make three distinct Chowla–Selberg periods — $\Gamma(1/4)$, $\Gamma(1/3)$, and $\Gamma(1/8)\Gamma(3/8)$ — visible as polygamma sums, and the full Eisenstein tower over both maximal-symmetry lattices is polygamma-explicit through weight 12. Third, *class field theory*: for discriminants of class number 2 through 5 ($-20, -15, -23, -39, -47$), cross-class ratios are algebraic numbers whose norms forecast the prime content of class-product period monomials, and the row-sum values recover the Hilbert class polynomials H_{-39} and H_{-47} outright — integer coefficients of 20 and more digits emerging from sums of polygamma values at quadratic-irrational complex points. A separate elementary layer (reflection composed with conjugation) explains MathWorld’s “surprising” FoxTrot digamma identity as one point of a two-parameter family. We also record the honest negatives: the components opaque to these mechanisms are Eichler-integral/quasi-modular data, and cross-class *sums* of weight-4 values are not Galois-clean.

1 Introduction

The starting question was deliberately open-ended: *are there interesting identities for polygamma values at non-real arguments — or for a single real or imaginary component of such values, even when the other component is opaque?*

The mechanism that answers it is elementary. For $a \in \mathbb{C}$ off the real axis and $s \geq 1$, splitting the bilateral sum at $m = 0$ and using $\psi^{(s)}(a) = (-1)^{s+1} s! \sum_{m \geq 0} (m+a)^{-s-1}$ gives

$$\sum_{m \in \mathbb{Z}} \frac{1}{(m+a)^{s+1}} = \frac{(-1)^{s+1} \psi^{(s)}(a) + \psi^{(s)}(1-a)}{s!} = \frac{\psi^{(s)}(a) + \psi^{(s)}(1-a)}{s!} \quad (s \text{ odd}). \quad (1)$$

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Thus *row-summing a lattice sum turns it into a polygamma sum*. For the weight- $2k$ Eisenstein series of $\mathbb{Z} + \mathbb{Z}\tau$ ($\text{Im } \tau > 0$, $k \geq 2$), summing first over m in each row n and noting that the rows $\pm n$ coincide (the power is even),

$$G_{2k}(\tau) = \sum'_{(m,n) \in \mathbb{Z}^2} \frac{1}{(m + n\tau)^{2k}} = 2\zeta(2k) + \frac{2}{(2k-1)!} \sum_{n \geq 1} \left[\psi^{(2k-1)}(n\tau) + \psi^{(2k-1)}(1 - n\tau) \right]. \quad (2)$$

Everything in this article is an exploitation of (2) at points τ where the left-hand side is known: the maximal-symmetry CM points $\tau = i$ and $\tau = \rho = e^{i\pi/3}$, the isogenous neighbour $2i$ and the discriminant -8 point $i\sqrt{2}$, and the reduced-form points of the imaginary-quadratic discriminants of class numbers 2–5.

Epistemic status. These are results of experimental mathematics, recorded in the PolyLog corpus of the Smithereens repository. Unless explicitly marked classical, each identity was discovered or confirmed by PSLQ over a deliberately prepared basis and verified with `mpmath` at the stated precision (residuals quoted per identity, typically 50–300 significant digits), with the project’s *canary* discipline guarding against PSLQ false positives: every candidate relation must continue to hold when re-evaluated at a substantially higher precision than the search precision. Polygamma values at complex a were computed via the Hurwitz-zeta shift $\psi^{(m)}(a) = (-1)^{m+1} m! (a^{-m-1} + \zeta(m+1, 1+a))$, which keeps the Hurwitz zeta in its numerically safe region. None of the new identities below currently carries a symbolic proof certificate; converting them into the repository’s certificate format is future work (the certificate program is the subject of a companion article).

2 The elementary layer: reflection composed with conjugation

Before any CM input, one family of components is fixed by symmetry alone. The Schwarz reflection $\psi^{(m)}(\bar{z}) = \overline{\psi^{(m)}(z)}$ turns the reflection law into a statement about a single real component, leaving the other component untouched.

Result 2.1 (vertical-line trigamma law). *On the line $\text{Re } z = \frac{1}{2}$ the reflection law alone evaluates the real part of the trigamma function at every height: for all real t ,*

$$\text{Re } \psi^{(1)}\left(\frac{1}{2} + it\right) = \frac{\pi^2}{2 \cosh^2(\pi t)}.$$

In particular $\text{Re } \psi^{(1)}\left(\frac{1+i}{2}\right) = \pi^2/(2 \cosh^2(\pi/2))$ [residual 0 at 50 dps]. Off the symmetric line, combining reflection, conjugation and the recurrence gives e.g.

$$\text{Re } \psi^{(1)}(i) = -\frac{1}{2} - \frac{\pi^2}{2 \sinh^2 \pi} \quad [\text{residual 0 at 50 dps}].$$

The proof of the first identity is one line: on $\text{Re } z = \frac{1}{2}$ one has $1 - z = \bar{z}$, so the reflection law $\psi^{(1)}(z) + \psi^{(1)}(1 - z) = \pi^2/\sin^2(\pi z)$ reads $2 \text{Re } \psi^{(1)}(z) = \pi^2/\sin^2(\pi z)$ there.

The imaginary parts on this line stay opaque — they are odd-row lattice data in the sense of Section 9.

2.1 The FoxTrot identity demystified

MathWorld's *Digamma Function* page [5] calls the following identity “surprising”; it arises from the FoxTrot series $\sum (-1)^{n-1} n^2 / (n^3 + 1)$, whose partial fractions hit the roots of $n^2 - n + 1$:

$$-\psi\left(\frac{\omega}{2}\right) - \psi\left(-\frac{\omega^2}{2}\right) + \psi\left(\frac{1+\omega}{2}\right) + \psi\left(\frac{1-\omega^2}{2}\right) = 2\pi \operatorname{sech}\left(\frac{\sqrt{3}\pi}{2}\right), \quad \omega = e^{i\pi/3}. \quad (3)$$

Law 2.2 (line law for digamma differences). *For every shift $c \in (0, 1)$, on the line $\operatorname{Re} z = (1-c)/2$,*

$$\operatorname{Re}[\psi(z+c) - \psi(z)] = \pi \operatorname{Re} \cot(\pi \bar{z}).$$

[verified for $c = \frac{1}{2}, \frac{1}{3}, \frac{2}{3}$ at generic heights t]

The mechanism is the same one-liner: on that line the shift satisfies $z+c = 1-\bar{z}$, so reflection composed with conjugation eliminates the digamma values, leaving an elementary $\sinh / \cos$ closed form. The FoxTrot identity (3) is the instance $c = \frac{1}{2}$, $z = \frac{1}{4} + i\frac{\sqrt{3}}{4}$: its four arguments pair into $2\operatorname{Re}[\psi(z+\frac{1}{2}) - \psi(z)]$, and on $\operatorname{Re} z = \frac{1}{4}$ one has $\pi \operatorname{Re} \cot(\pi \bar{z}) = \pi \operatorname{sech}(2\pi t)$. The identity is thus the point $t = \sqrt{3}/4$ on a line of identities, itself one member of a two-parameter family — no complex multiplication required.

Differentiating in t ladders the law up the polygamma tower [weights 2, 3 verified at 10^{-50}]:

$$\begin{aligned} \operatorname{Im}[\psi^{(1)}(\tfrac{3}{4} + it) - \psi^{(1)}(\tfrac{1}{4} + it)] &= 2\pi^2 \operatorname{sech}(2\pi t) \tanh(2\pi t), \\ \operatorname{Re}[\psi^{(2)}(\tfrac{3}{4} + it) - \psi^{(2)}(\tfrac{1}{4} + it)] &= 8\pi^3 \operatorname{sech}^3(2\pi t) - 4\pi^3 \operatorname{sech}(2\pi t). \end{aligned}$$

At each weight the alternating component (Re at odd weight, Im at even weight) stays opaque, exactly as in Section 9.

3 Maximal symmetry: vanishing identities and Γ -periods

At the two CM points with extra symmetry — $\tau = i$ (square lattice, $j = 1728$) and $\tau = \rho = e^{i\pi/3}$ (hexagonal lattice, $j = 0$) — classical theory pins every G_{2k} exactly. The square lattice admits no weight-6 form ($i \cdot (\mathbb{Z} + \mathbb{Z}i) = \mathbb{Z} + \mathbb{Z}i$ forces $G_6(i) = i^{-6}G_6(i) = -G_6(i)$), and the hexagonal lattice no weight-4 form; $G_4(i)$ and $G_6(\rho)$ are the lemniscatic and equianharmonic period powers — monomials in $\Gamma(1/4)$, resp. $\Gamma(1/3)$, by Chowla–Selberg [1].

3.1 Vanishing identities

Result 3.1 (the CM gems).

$$\begin{aligned} \sum_{n \geq 1} [\psi^{(5)}(ni) + \psi^{(5)}(1-ni)] &= -120 \zeta(6) \quad [\text{residual } 1.7 \times 10^{-49}], \\ \sum_{n \geq 1} [\psi^{(3)}(n\rho) + \psi^{(3)}(1-n\rho)] &= -6 \zeta(4), \quad \rho = e^{i\pi/3} \quad [\text{residual } 6.8 \times 10^{-50}]. \end{aligned}$$

Each individual bracket is a transcendental-looking complex number (elementary in $\coth / \operatorname{csch}$ of $\pi n \sqrt{3}/2$ and the like after the bilateral cotangent expansion, but nobody would guess the totals); the sums collapse to rational multiples of $\zeta(6)$, $\zeta(4)$ *because the corresponding modular forms do not exist*. Via (2) these are exactly the statements $G_6(i) = 0$ and $G_4(\rho) = 0$.

3.2 Chowla–Selberg made polygamma-visible

Result 3.2 (period identities).

$$2\zeta(4) + \frac{1}{3} \sum_{n \geq 1} [\psi^{(3)}(ni) + \psi^{(3)}(1 - ni)] = \frac{\Gamma(1/4)^8}{960 \pi^2} \quad [\text{residual } 2.1 \times 10^{-50}],$$

$$2\zeta(6) + \frac{1}{60} \sum_{n \geq 1} [\psi^{(5)}(n\rho) + \psi^{(5)}(1 - n\rho)] = \frac{\Gamma(1/3)^{18}}{8960 \pi^6} \quad [\text{residual } 2.0 \times 10^{-60}].$$

Polygamma values at purely imaginary (resp. hexagonal) points, summed over the integer multiples of the point, know the lemniscate constant and its equianharmonic sibling. The atoms $\Gamma(1/4)^2/\pi$ and $\Gamma(1/3)$ are precisely the CM-period currency of the repository’s hypergeometric reducer (`cmPeriodReduce` in `src/Hypergeometric`), so these identities tie the polygamma world directly to that corpus.

3.3 The complete low-weight table

The symmetry argument and the classical Hurwitz recursion [3]

$$(2n+1)(2n-1)(n-3)G_{2n} = 3 \sum_{m=2}^{n-2} (2m-1)(2n-2m-1)G_{2m}G_{2n-2m}$$

extend both subsections through weight 12 with no new constants [all entries verified, residuals 10^{-60} – 10^{-263}]:

weight $2k$	square lattice $\tau = i$	hexagonal lattice $\tau = \rho$
4	$\Gamma(1/4)^8/(960\pi^2)$	0
6	0	$\Gamma(1/3)^{18}/(8960\pi^6)$
8	$\frac{3}{7}G_4(i)^2 = 3\Gamma(1/4)^{16}/(7 \cdot 960^2\pi^4)$	0
10	0	0
12	$\frac{1}{143}(18G_4(i)^3 + 25G_6(i)^2)$ type	$\frac{25}{143}G_6(\rho)^2 \propto \Gamma(1/3)^{36}$

Every bold zero is a vanishing identity $\sum_{n \geq 1} [\psi^{(w-1)}(n\tau) + \psi^{(w-1)}(1 - n\tau)] = -(w-1)! \zeta(w)$, and every non-zero entry is a $\psi^{(w-1)}$ -sum evaluation. The pattern is exhaustive: $G_{2k}(i) = 0$ iff $2k \not\equiv 0 \pmod{4}$, $G_{2k}(\rho) = 0$ iff $2k \not\equiv 0 \pmod{6}$, and all non-zero values are $\mathbb{Q}$ -monomials in the two period powers — so the *entire* Eisenstein tower over both lattices is polygamma-explicit.

4 Beyond maximal symmetry: $2i$ and $i\sqrt{2}$

The same row-sum at the next two CM points, with closed forms found by PSLQ [120-dps canaries]:

Result 4.1.

$$G_4(2i) = \frac{11}{15360} \frac{\Gamma(1/4)^8}{\pi^2} \quad [\text{canary } 0 \text{ at } 120 \text{ dps}],$$

$$G_4(i\sqrt{2}) = \frac{\Gamma(1/8)^4 \Gamma(3/8)^4}{4608 \pi^2} \quad [\text{canary } 3.9 \times 10^{-121}].$$

The weight-6 row at the same points [160-dps canaries]:

$$G_6(2i) = \frac{\Gamma(1/4)^{12}}{2^{14} \cdot 5 \cdot \pi^3}, \quad G_6(i\sqrt{2}) = \frac{\Gamma(1/8)^6 \Gamma(3/8)^6}{2^{12} \cdot 3^3 \cdot 5 \cdot \pi^3}.$$

The first value confirms the isogeny picture: $2i$ sits over the same field $\mathbb{Q}(i)$, so only the rational factor changes ($G_4(i)/G_4(2i) = 16/11$). The second imports a genuinely *new period*: discriminant -8 , field $\mathbb{Q}(\sqrt{-2})$ — the polygamma sum $\sum_n [\psi^{(3)}(ni\sqrt{2}) + \psi^{(3)}(1 - ni\sqrt{2})]$ evaluates in $\Gamma(1/8)\Gamma(3/8)$. The fundamental unit $1 + \sqrt{2}$ was offered in the PSLQ basis and declined (exponent 0). Together with Section 3 this makes three distinct Chowla–Selberg periods polygamma-visible.

5 Class number 2: discriminants -20 and -15

At $\mathbb{Q}(\sqrt{-5})$ (class number 2) a single-value logarithmic PSLQ *must* fail: the algebraic factor in G_4/Ω^4 lives in the Hilbert class field and is not a monomial. The structure that works is the Hecke/Damerell pair [2] over the two reduced forms $x^2 + 5y^2$ ($\tau_1 = i\sqrt{5}$) and $2x^2 + 2xy + 3y^2$ ($\tau_2 = (-1 + i\sqrt{5})/2$).

Result 5.1 (disc -20).

$$\frac{G_4(\tau_1)}{G_4(\tau_2)} = \frac{29 + 12\sqrt{5}}{44} = \frac{(3 + 2\sqrt{5})^2}{44} \quad [\text{canary 0 at 160 dps}],$$

$$G_4(\tau_1) G_4(\tau_2) = \frac{11 \varphi^4 \Gamma(3/20)^8 \Gamma(7/20)^8}{2^{14} \cdot 3^4 \cdot 5^4 \cdot \pi^4} \quad [\text{canary } 5.3 \times 10^{-160}],$$

where φ is the golden ratio. The ratio is algebraic of degree 2 with norm $N(29 + 12\sqrt{5}) = 121 = 11^2$; the class product is a period monomial carrying the prime 11 and a golden unit.

Result 5.2 (disc -15). With $\tau_1 = (1 + i\sqrt{15})/2$ for $x^2 + xy + 4y^2$ and the conjugate pair $\tau_{2,3} = (\pm 1 + i\sqrt{15})/4$ representing $2x^2 + xy + 2y^2$, write $\widehat{G}_4 = \frac{1}{2}[G_4(\tau_2) + G_4(\tau_3)] = \text{Re } G_4(\tau_2)$ for the class-pair average. Then [160-dps canaries]:

$$\frac{G_4(\tau_1)}{\widehat{G}_4} = \frac{2(4 + \sqrt{5})^2}{77}, \quad G_4(\tau_1) \widehat{G}_4 = \frac{77 \left(\Gamma(\frac{1}{15})\Gamma(\frac{2}{15})\Gamma(\frac{4}{15})\Gamma(\frac{8}{15})\right)^4}{2^{13} \cdot 3^6 \cdot 5^5 \cdot \pi^4}.$$

Unlike disc -20 , the non-principal value here is genuinely complex (the form $(2, 1, 2)$ has $|\tau_2| = 1$, where conjugation acts nontrivially on G_4); the class-pair average is the Galois-natural object.

The Γ -block in Result 5.2 is exactly the set of residues $\{1, 2, 4, 8\}$ with $\chi_{-15} = +1$; the golden unit is declined by PSLQ (exponent 0); and the prime content $77 = 7 \cdot 11$ of the product matches the ratio's denominator ($N(4 + \sqrt{5}) = 11$). The heuristic that emerges, used at every later discriminant: *run the ratio first; its norm names the primes the class-product basis needs.*

6 Class number 3: discriminant -23

$\mathbb{Q}(\sqrt{-23})$ has $h = 3$, with Hilbert class field generated by the famous cubic $x^3 - x - 1$. The three reduced forms give $\tau_1 = (1 + i\sqrt{23})/2$ and $\tau_{2,3} = (\pm 1 + i\sqrt{23})/4$; $G_4(\tau_1)$ is real and $G_4(\tau_{2,3})$ form a conjugate pair.

Result 6.1 (disc -23 class product). [*canary* 1.4×10^{-160}]

$$G_4(\tau_1) G_4(\tau_2) G_4(\tau_3) = \frac{11 \cdot 17 P^4}{2^{24} \cdot 3^6 \cdot 23^5 \cdot \pi^{16}}, \quad P = \prod_{\substack{k \bmod 23 \\ (\frac{k}{23})=+1}} \Gamma\left(\frac{k}{23}\right) \quad (11 \text{ factors}).$$

Result 6.2 (the γ_2 class polynomial from row-sums). *With $\gamma_2 = E_4/\eta^8$ phase-corrected by the canonical cube root (the $q^{1/3}$ ambiguity of η^8 at these points), the three values $\gamma_2(\tau_i)$ computed from the polygamma row-sums are exactly the roots of the classical class polynomial*

$$x^3 + 155x^2 + 650x + 23375 = 0 \quad [\text{coefficients integer to } 10^{-80}],$$

whose constant term $23375 = 5^3 \cdot 11 \cdot 17$ forecast the primes 11, 17 in the class product — the norm heuristic of Section 5, one degree up.

Negative result 6.3 (weight-4 sums are not Galois-clean). *The dimensionless symmetric functions e_1^3/e_3 and e_2^3/e_3^2 of the raw G_4 values are not rational (PSLQ to coefficient height 10^{30} at 300 dps). Unlike the value product (a norm) and the weight-0 γ_2 invariants, sums of weight-4 CM values across ideal classes are not Galois-stable — the period correction under $\text{Gal}(H/K)$ is nontrivial at weight 4. Class polynomials must be formed from $\gamma_2 = E_4/\eta^8$ (or j), not from the G_4 values themselves.*

7 Class number 4: discriminant -39 and genus structure

The four reduced forms of discriminant -39 , namely $(1, 1, 10)$, $(2, \pm 1, 5)$, $(3, 3, 4)$, give $\tau \in \{(1 + i\sqrt{39})/2, (\pm 1 + i\sqrt{39})/4, (3 + i\sqrt{39})/6\}$.

Result 7.1 (disc -39 class product). [*160-dps canary*]

$$\prod_{i=1}^4 G_4(\tau_i) = \frac{17 \cdot 23 \cdot 29 P^4}{2^{28} \cdot 3^9 \cdot 5^4 \cdot 13^8 \cdot \pi^{16}}, \quad P = \prod_{\substack{k \bmod 39 \\ \chi_{-39}(k)=+1}} \Gamma\left(\frac{k}{39}\right) \quad (12 \text{ factors}).$$

A wrinkle relative to Section 6: γ_2 is *not* a class invariant here ($3 \mid 39$, the classical obstruction [4]; observed directly — no cube-root phase choice makes the γ_2 -sums rational). But j always is, and the row-sum values deliver the Hilbert class polynomial outright.

Result 7.2 (Hilbert class polynomial of disc -39).

$$H_{-39}(x) = x^4 + 331531596 x^3 - 429878960946 x^2 + 109873509788637459 x + 20919104368024767633$$

[$e_1, \dots, e_3$ PSLQ-exact integers; e_4 integral to 10^{-137}].

Worth dwelling on: a quartic with 20-digit integer coefficients recovered from sums of polygamma values at quadratic-irrational complex points. The polygamma row-sum is a perfectly serviceable special-function front door to the classical CM tower.

Result 7.3 (genus structure inside the quartet). *The class group is $\mathbb{Z}/4$, with the ambiguous form $(3, 3, 4)$ as the order-2 class. The raw cross-class ratio v_1/v_4 is honestly quartic (no degree-2 minimal polynomial to height 10^{12}), but the genus-wise ratio — principal genus $\{1, A^2\}$ over*

$\{A, A^3\}$ — is exactly quadratic, in the real part of the genus field $\mathbb{Q}(\sqrt{-3}, \sqrt{13})$ as genus theory demands [160-dps canary; $\sqrt{3}, \sqrt{39}$ declined by PSLQ]:

$$\frac{G_4(\tau_1) G_4(\tau_4)}{G_4(\tau_2) G_4(\tau_3)} = \frac{103299 + 4440\sqrt{13}}{181424}, \quad N = \left(\frac{9}{16}\right)^2, \quad 181424 = 16 \cdot 17 \cdot 23 \cdot 29.$$

The denominator reuses the class-product primes 17, 23, 29.

8 Class number 5: discriminant -47

The five reduced forms $(1, 1, 12)$, $(2, \pm 1, 6)$, $(3, \pm 1, 4)$ give the quintet $\tau \in \{(1 + i\sqrt{47})/2, (\pm 1 + i\sqrt{47})/4, (\pm 1 + i\sqrt{47})/6\}$.

Result 8.1 (disc -47 class product). [200-dps canary]

$$\prod_{i=1}^5 G_4(\tau_i) = \frac{11^2 \cdot 23 \cdot 29 P^4}{2^{52} \cdot 3^6 \cdot 47^9 \cdot \pi^{36}}, \quad P = \prod_{\substack{k \bmod 47 \\ \left(\frac{k}{47}\right)=+1}} \Gamma\left(\frac{k}{47}\right) \quad (23 \text{ factors}).$$

Result 8.2 (Hilbert class polynomial of disc -47). The row-sum j -values deliver the quintic Hilbert class polynomial [all symmetric functions integral to 10^{-129} or better]:

$$H_{-47}(x) = x^5 + 2257834125 x^4 - 9987963828125 x^3 + 5115161850595703125 x^2 - 14982472850828613281250 x + 16042929600623870849609375.$$

The CM tower is now polygamma-explicit through class number 5 with one uniform recipe: row-sum $\rightarrow$ class-product log-PSLQ (norm-forecast primes) $\rightarrow$ j -symmetric-function integrality.

9 The opaque components

The components *not* fixed by reflection-conjugation or by the even-row mechanism — $\text{Im } \psi^{(1)}(i)$, $\text{Re } \psi(i)$ beyond γ , $\text{Im } \psi^{(1)}\left(\frac{1+i}{2}\right)$ — are *odd-row* lattice data: for example

$$\text{Im } \psi^{(1)}(i) = -2 \sum_{k \geq 1} \frac{k}{(k^2 + 1)^2}$$

is an odd-power analogue of the coth family, which the bilateral (even) cotangent expansion cannot reach. These are Eichler-integral/quasi-modular territory — the E_2 -flavoured layer; compare $\sum_{k \geq 1} k/(e^{2\pi k} - 1) = \frac{1}{24} - \frac{1}{8\pi}$, which is $E_2(i) = 3/\pi$ in Lambert-series clothing.

Negative result 9.1. A PSLQ basket for these constants against products drawn from

$$\{1, \pi, \coth \pi, \pi^2/\sinh^2 \pi, \pi \coth \pi, \dots\}$$

found nothing at small coefficient heights — consistent with genuinely new constants. Recorded as an honest negative; the Lambert-series face ($\sum \sigma_5(n)q^n$ at $q = e^{-2\pi}$ and $q = -e^{-\pi\sqrt{3}}$) is the natural next probe surface.

10 Methodology notes

Two failure chains from the discovery process are worth recording.

A vanishing identity passing is weak evidence for the machinery around it. The row-sum prefactor in (2) is $2/(2k-1)!$; an early weight-6 probe used $1/120$ instead of $1/60$. The *vanishing* tests are blind to such a factor (zero swallows it) — they passed while the value probe quietly computed garbage. The bug was caught by cross-checking the lattice row-sum against the modular q -series $G_6 = 2\zeta(6)E_6(\rho)$ with $q = -e^{-\pi\sqrt{3}}$ (agreement to 30+ digits).

When a CM-value PSLQ fails, anchor on η . With the correct value in hand, log-PSLQ over $\{\ln \Gamma(1/3), \ln \pi, \ln 2, \ln 3\}$ still found nothing for $G_6(\rho)$ — a span failure, not a precision failure: $8960 = 2^8 \cdot 5 \cdot 7$ and $\zeta(6) = \pi^6/945$ drag in the primes 5 and 7. The debugging anchor was the Dedekind eta function: PSLQ over the same small basis instantly returned the classical $|\eta(\rho)| = 3^{1/8}\Gamma(1/3)^{3/2}/(2\pi)$, and then $E_6(\rho)^2 = 1728|\eta(\rho)|^{24}$ forces the closed form with no further search (note $\eta(\rho)^{24} = \Delta(\rho)$ is *negative* real here, since $q = -e^{-\pi\sqrt{3}}$ and $E_4(\rho) = 0$). Eta’s closed forms have tiny prime support; derive upward through the modular algebra rather than widening a PSLQ basis blindly.

11 Open directions

1. The G_6 row at the class-number ≥ 2 points, and the finer Galois structure inside the disc -39 quartet (which pairs of G_4 values generate which subfield of the class field).
2. The quasi-modular layer: $E_2(i) = 3/\pi$ row-summed is weight 2 — conditionally convergent rows requiring Eisenstein summation — and should explain exactly which combinations of the opaque odd constants of Section 9 are reachable.
3. Folding the period identities into the repository’s identity stores once a store schema for infinite-sum claims exists (current stores hold finite reduction rules only), and converting the finite-rank discoveries into multiplier proof certificates.

Sources and verification artifacts

This article presents results recorded in the Smithereens repository (`src/PolyLog`), principally in the report `docs/reports/polygamma-complex-arguments-cm-lattices__d41f7be20c61.md` and the commit range `272af472..68236dab4` (2026-06-09 through 2026-06-11). All numerical verification used `mpmath` at the per-identity precisions quoted in square brackets; PSLQ discoveries were re-verified at elevated precision (canary discipline). The classical ingredients — the Chowla–Selberg formula, the structure of $M_*(\mathrm{SL}_2(\mathbb{Z}))$, Hurwitz’s recursion, genus theory, and the class polynomials of discriminants -23 , -39 , -47 — are classical; the polygamma-explicit formulations and the specific ratio/class-product evaluations are, to the project’s knowledge, newly recorded here.

References

- [1] A. Selberg and S. Chowla, *On Epstein's zeta-function*, J. reine angew. Math. **227** (1967), 86–110.
- [2] R. M. Damerell, *L-functions of elliptic curves with complex multiplication, I*, Acta Arith. **17** (1970), 287–301.
- [3] A. Hurwitz, *Über die Entwicklungskoeffizienten der lemniskatischen Funktionen*, Math. Ann. **51** (1899), 196–226.
- [4] D. A. Cox, *Primes of the Form $x^2 + ny^2$: Fermat, Class Field Theory, and Complex Multiplication*, Wiley, 1989. (Genus theory; class invariants; the $3 \mid D$ obstruction for γ_2 .)
- [5] E. W. Weisstein, *Digamma Function*, MathWorld — A Wolfram Web Resource. (Source of the FoxTrot identity.)